PROJECT WORK REPORT

FEM Simulation of PCF-SPR Biosensors for Cancer Cell Detection

Photonics; Numerical Verification of Photonic Crystal Fiber / Surface Plasmon Resonance Sensor Designs

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Simulation Tool	COMSOL Multiphysics; Finite Element Method (FEM), wave-optics mode analysis

1. Scope of This Report

The full seminar deliverable was a literature review on photonic crystal fiber (PCF) biosensors for cancer detection. This report isolates only the original, hands-on portion of that work: two FEM simulations independently built and run in COMSOL Multiphysics to numerically verify the reported sensing performance of two published PCF-SPR (surface plasmon resonance) biosensor designs. Background theory and the literature survey are omitted here to keep the focus on the engineer's own simulation work.

2. Objective

- Reconstruct two published PCF-SPR biosensor cross-sections as parametric FEM geometries in COMSOL.
- Run wave-optics (mode) analysis to extract the effective mode index, confinement loss, and core/SPP mode field distributions as a function of wavelength.
- Quantify sensor performance; wavelength sensitivity, amplitude sensitivity, resolution, and figure of merit (FOM); from the simulated loss and effective-index spectra.

3. Simulation 1; V-Shaped Gold-Coated SPR-PCF Sensor

A V-notch surface-groove PCF-SPR sensor was modeled and solved by finite element mode analysis in COMSOL. The gold plasmonic layer (highlighted in the geometry render) sits in the V-shaped groove on the flattened fiber face, positioned to maximize coupling between the core-guided mode and the surface plasmon polariton (SPP) mode.

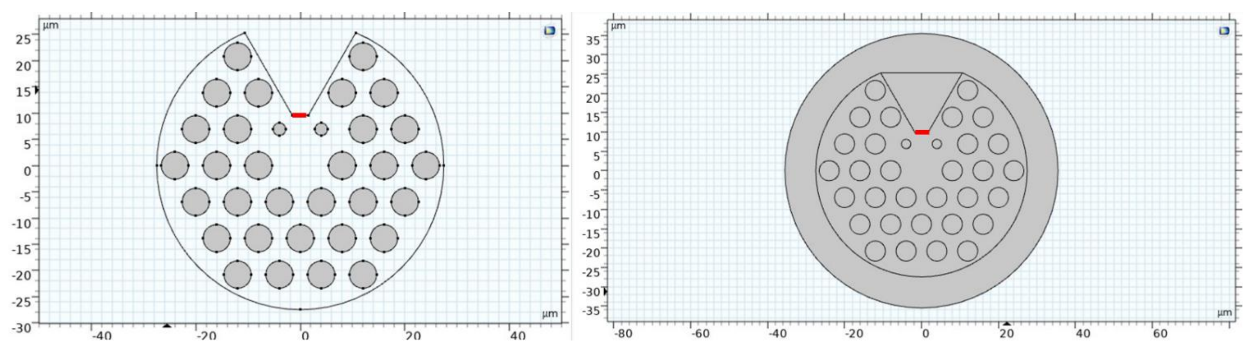

Figure 1; COMSOL cross-section geometry of the V-shaped SPR-PCF sensor (gold layer highlighted)

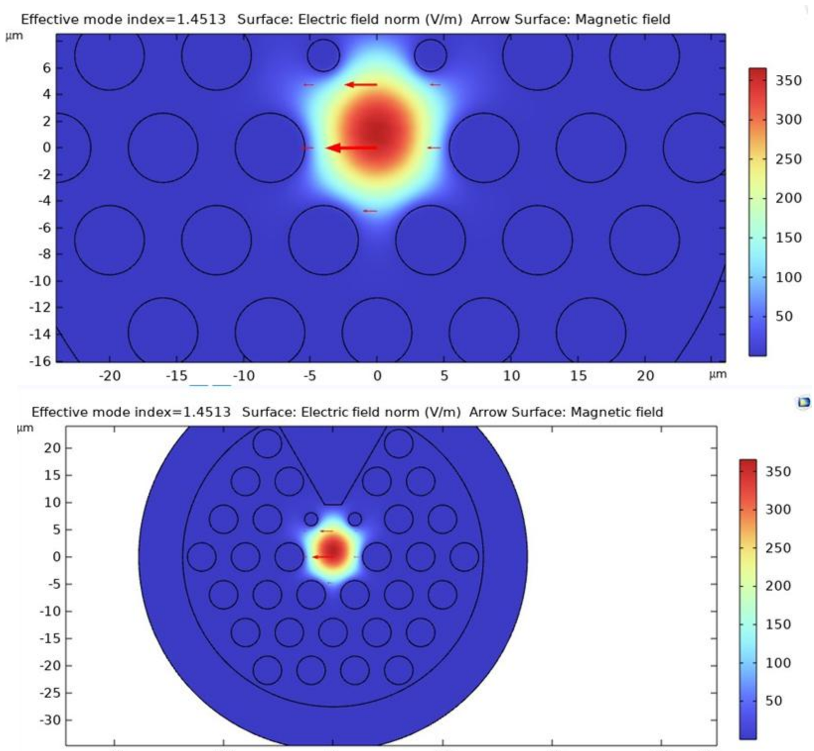

Figure 2; Simulated electric-field norm and magnetic-field vector distribution of the core-guided mode (effective mode index = 1.4513)

Key simulated performance figures extracted from the wavelength-scan (loss vs. wavelength, effective index vs. wavelength) results:

Metric	Simulated Value
Analyte refractive-index range	1.29 – 1.44
Wavelength sensitivity	88,000 nm/RIU
Amplitude sensitivity	3136 RIU^{-1}
Resolution	1.136×10^{-6} RIU
Figure of Merit (FOM)	1517

A geometric-tolerance sweep (pitch, air-hole diameter, channel size) was also run to confirm the design's stability against small fabrication variations.

4. Simulation 2; Gold-Coated PCF-SPR Sensor for Ultrasensitive RI Detection

A second PCF-SPR sensor, using gold as the plasmonic coating over a wider refractive-index detection window, was modeled and solved the same way; FEM wave-optics mode analysis in COMSOL; sweeping wavelength to extract loss and effective-index spectra for both TE- and TM-like mode families.

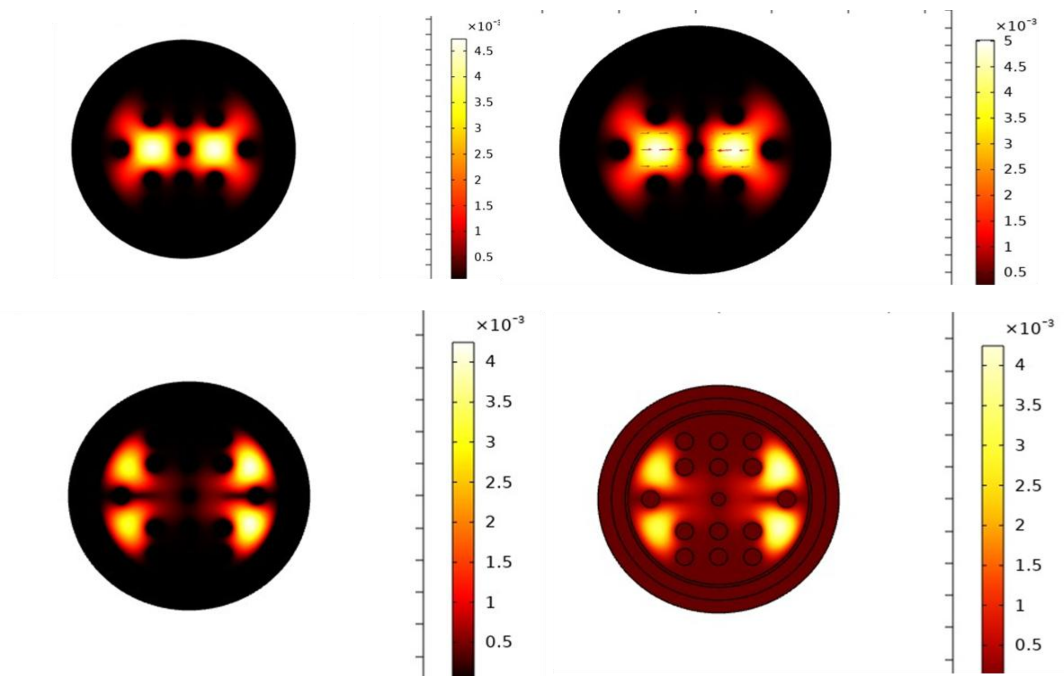

Figure 3; Simulated modal electric-field distributions across the wavelength sweep for the gold-coated SPR-PCF sensor

Key simulated performance figures:

Metric	Simulated Value
Analyte refractive-index range	1.27 – 1.43
Wavelength sensitivity	157,000 nm/RIU (resolution 6.37×10^{-7} RIU)
Amplitude sensitivity	1263 RIU^{-1} (resolution 7.92×10^{-5} RIU)
Figure of Merit (FOM)	1648

5. Skills Demonstrated

- Finite Element Method (FEM) simulation of guided-wave and plasmonic modes in COMSOL Multiphysics.
- Parametric CAD reconstruction of photonic crystal fiber cross-sections (air-hole lattices, plasmonic coatings, analyte channels) from published geometries.
- Mode analysis: effective mode index extraction, confinement-loss spectra, and core–SPP phase-matching identification.
- Sensor performance quantification; wavelength sensitivity, amplitude sensitivity, resolution, and figure of merit; from simulated spectral data.
- Design robustness assessment via geometric-tolerance (pitch/diameter) sweeps.

6. Outcome

Both independently built COMSOL models reproduced sensing performance consistent with the source publications, confirming correct implementation of the PCF-SPR mode-analysis workflow and providing a validated simulation basis for further, self-directed PCF-SPR sensor design work.